

ISIT, Russian Federation

WMO: 249510

Lat: 60.817N

Lon: 125.317E

Elev: 131

StdP: 99.77

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -46.3	(c) -44.6	(d) -48.6	(e) 0.0	(f) -46.3	(g) -47.0	(h) 0.0	(i) -44.6	(j) 8.6	(k) -29.3	(l) 8.1	(m) -29.7	(n) 0.7	(o) 250	(p) 0.671

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.7	28.4	19.8	26.4	18.8	24.4	17.8	21.0	26.3	19.8	24.8	18.6	23.1	2.0	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.1	14.1	24.0	17.9	13.0	22.5	16.7	12.1	21.5	61.7	26.4	57.5	25.0	53.6	23.1	25.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 7.5	(b) 6.5	(c) 5.5	(e) -47.8	(f) 32.4	(g) 2.3	(h) 1.7	(i) -49.4	(j) 33.7	(k) -50.8	(l) 34.7	(m) -52.0	(n) 35.6	(o) -53.7	(p) 36.9		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	-6.7	-33.3	-29.3	-18.2	-3.8	7.0	15.4	18.4	15.0	6.7	-4.8	-22.0	-32.5	(6)
(7)		DBStd	19.75	7.97	7.35	8.01	5.85	4.10	3.99	3.61	3.66	3.92	5.95	8.79	7.55	(7)
(8)		HDD10.0	6702	1342	1101	874	413	111	6	1	4	113	460	961	1317	(8)
(9)		HDD18.3	9197	1600	1334	1132	663	351	103	43	115	350	718	1211	1575	(9)
(10)		CDD10.0	620	0	0	0	0	18	168	261	160	13	0	0	0	(10)
(11)		CDD18.3	73	0	0	0	0	0	16	46	12	0	0	0	0	(11)
(12)		CDH23.3	647	0	0	0	0	2	151	391	102	1	0	0	0	(12)
(13)		CDH26.7	151	0	0	0	0	0	28	105	17	0	0	0	0	(13)
(14)	Wind	WSAvg	1.8	1.9	1.7	1.5	1.6	2.0	1.7	1.7	2.0	1.9	1.9	1.9		(14)
(15)	Precipitation	PrecAvg	290	12	8	8	9	22	44	53	36	20	15	13		(15)
(16)		PrecMax	458	25	14	21	26	88	87	98	104	100	32	26	24	(16)
(17)		PrecMin	186	3	2	2	1	2	7	4	2	15	5	5	5	(17)
(18)		PrecStd	69	6	3	5	6	17	22	21	31	20	9	6	5	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-8.7	-6.4	4.7	13.2	23.0	29.6	32.3	28.7	20.3	9.0	1.5	-7.9	(19)
(20)			MCWB	-9.3	-7.3	0.3	6.2	13.0	18.5	21.7	20.3	13.4	4.8	0.1	-8.5	(20)
(21)		2%	DB	-14.5	-10.8	0.1	9.4	19.7	27.0	29.5	26.1	17.1	6.0	-1.5	-14.1	(21)
(22)			MCWB	-15.0	-11.5	-2.5	3.8	11.1	17.7	20.5	18.7	11.9	3.7	-2.3	-14.5	(22)
(23)		5%	DB	-17.8	-14.9	-2.6	6.7	17.0	25.0	27.4	23.9	14.8	4.0	-5.6	-18.1	(23)
(24)			MCWB	-18.2	-15.5	-4.5	2.2	9.7	16.9	19.7	17.9	10.7	2.0	-6.7	-18.5	(24)
(25)		10%	DB	-21.5	-18.5	-5.7	4.7	14.6	22.8	25.4	21.8	12.9	2.4	-9.5	-21.7	(25)
(26)			MCWB	-21.8	-18.9	-7.1	1.0	8.5	15.8	18.9	16.6	9.8	0.7	-10.2	-21.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-9.2	-7.2	0.5	6.4	13.8	20.6	23.1	21.5	14.4	5.9	0.3	-8.3	(27)
(28)			MCDWB	-8.5	-6.6	4.1	12.8	21.3	27.1	29.4	26.9	18.4	7.6	1.2	-7.8	(28)
(29)		2%	WB	-15.0	-11.6	-2.3	4.2	11.7	18.9	21.7	19.8	12.8	3.9	-2.4	-14.5	(29)
(30)			MCDWB	-14.6	-10.9	0.1	8.8	18.1	24.8	27.6	24.3	16.1	5.8	-1.5	-14.1	(30)
(31)		5%	WB	-18.2	-15.5	-4.5	2.6	10.3	17.7	20.7	18.5	11.4	2.2	-6.6	-18.5	(31)
(32)			MCDWB	-17.8	-14.9	-2.6	6.0	16.0	23.3	25.9	22.6	14.0	3.7	-5.7	-18.1	(32)
(33)		10%	WB	-21.8	-18.9	-7.1	1.3	8.9	16.6	19.6	17.3	10.1	0.7	-10.3	-22.0	(33)
(34)			MCDWB	-21.5	-18.5	-5.7	4.3	13.8	21.6	24.4	21.1	12.6	2.2	-9.5	-21.8	(34)
(35)	Mean Daily Temperature Range	MDBR		6.2	8.0	12.2	12.3	10.4	11.3	10.7	10.1	8.0	6.0	6.2	5.7	(35)
(36)		5% DB	MCDBR	9.2	10.5	12.6	12.8	14.9	14.6	13.9	13.9	12.0	6.7	9.4	9.8	(36)
(37)			MCWBR	8.8	9.9	10.5	8.2	8.2	7.1	6.9	7.8	7.6	5.2	9.1	9.6	(37)
(38)		5% WB	MCDWB	9.1	10.5	12.4	11.3	13.6	12.6	11.9	11.8	10.6	6.3	9.2	9.7	(38)
(39)			MCWBR	8.8	10.0	10.4	7.4	7.8	6.7	6.6	7.2	7.1	5.2	8.9	9.5	(39)
(40)	Clear-Sky Solar Irradiance	taub		0.239	0.274	0.308	0.365	0.427	0.427	0.516	0.464	0.351	0.295	0.240	0.210	(40)
(41)		taud		1.965	2.039	2.052	2.038	2.125	2.241	1.976	2.089	2.392	2.305	2.002	1.892	(41)
(42)		Ebn at Noon		471	692	797	816	792	802	705	712	763	680	462	332	(42)
(43)		Edh at Noon		45	79	110	137	137	124	159	131	81	61	42	29	(43)
(44)	All-Sky Solar Radiation	RadAvg		0.45	1.48	3.20	5.08	5.29	5.76	5.22	4.00	2.47	1.26	0.61	0.24	(44)
(45)		RadStd		0.03	0.07	0.12	0.18	0.34	0.43	0.41	0.26	0.24	0.15	0.05	0.01	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.80	N/A	N/A	N/A	N/A	N/A	-271	-278	N/A	N/A
(47) Regional (3 neighbors)		+0.80	N/A	N/A	N/A	N/A	N/A	-271	-278	N/A	N/A

Nomenclature: See separate page